

MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap

Oct 6 – 8 | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct a 95% confidence interval for a mean or proportion as statistic $\pm 2 \cdot \text{SE}$, given the statistic and its standard error. (CLO 6) 2. Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CLO 4, CLO 6)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals Interactive Lecture: 3.2 Understanding and Interpreting Confidence Intervals Slides: 3.2 Understanding and Interpreting Confidence Intervals	Participation: Interpreting Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website) Zoom (live class)
3. Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CLO 5, CLO 6) 4. Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CLO 5, CLO 6) 5. Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)	e-Text: 3.3 Constructing Bootstrap Confidence Intervals Interactive Lecture: 3.3 Constructing Bootstrap Confidence Intervals Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator) StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	Participation: Bootstrap a Proportion – Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI	Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website) Zoom (live class)
<i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i>			Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 4 Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals.